



Unity University
Faculty of Computational Sciences
and Software Engineering

Local Event Hub – Android Application

MAD Programming Project

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Engineering**

Department of Computer Science

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Members :	ID
1. Eyob Medihin	UU93170R
2. Leul Yeshitila	UU92774R
3. Meklit Yikuno	UU92852R
4. Nathan Abiy	UU92863R
5. Robel Shewangzaw	UU92779R

1. Introduction

The Local Event Hub is an Android mobile application developed to help users discover, create, and manage local events happening across Addis Ababa. It provides a centralized platform where users, organizers, and admins can interact through a role-based system.

The application is developed using:

- Java (backend logic)
- XML (UI layouts)
- Room Database (offline storage)
- Android Studio (IDE)
- MVVM architecture

The system allows ordinary users to browse events, organizers to create events, and administrators to approve or reject event submissions.

2. Purpose of the System

The purpose of the Local Event Hub is to:

- Allow residents to easily discover nearby events.
 - Provide organizers with a platform to publish their events.
 - Maintain event quality through admin approval.
 - Offer district-based event search (Bole, Arada, Yeka, Nifas Silk).
 - Improve access to local activities, concerts, workshops, etc.
 - Reduce the difficulty of manually searching for event information.
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3. Key Features

✓ User Registration & Login

Users can register and log in using their credentials.

✓ Role-Based Access

- **User** – view events, search by district
- **Organizer** – create/manage events
- **Admin** – approve or reject events

✓ Event Management

- Create event
- Edit event
- Delete event
- Approve/reject (admin)

✓ District Filtering

Search events by location.

✓ Favorites

Users can bookmark events.

✓ Modern UI

Uses Material Design components and supports Dark Mode.

3.1 Functional Requirements

User Registration

- Users can register with a username, password, and role.
- Details are stored in Room Database.

User Login

- Valid credentials required to access features.
- Incorrect login displays an error message.

Event Creation (Organizer)

Event details include:

- Title
- Description
- Date
- Location
- District
- Organizer ID

Event Approval (Admin)

- Admin views all event requests.
- Admin can approve or reject events.

Event Listing (User)

- List shows only approved events.
- Events displayed using RecyclerView.

District Filtering

Users search events by districts such as:

- Bole
- Arada
- Yeka
- Nifas Silk

Favorite Management

Users can mark/unmark events as favorite.

Navigation

Home page displays options based on user role.

4. Non-Functional Requirements

Usability

- Simple UI and easy navigation
- Clear buttons and labels

Performance

- Event list loads quickly
- Searches return results within seconds

Compatibility

- Android 5.0 (API 21) and above

- Works on phones and tablets

Reliability

- Event data stored safely in Room
- No data loss on restart

Security

- Basic authentication
- Only admin can approve events

Maintainability

- MVVM structure makes code easy to update

Scalability

Future possible extensions:

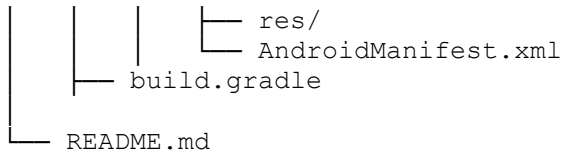
- Maps
- Push notifications
- Cloud database

System Constraints

- Works offline (local database only)
- No cloud syncing
- Basic authentication (no OAuth)

5. Project Structure Overview

```
LocalEventHub/
├── app/
│   └── src/
│       └── main/
│           └── java/com/example/localeventhub/
│               ├── activities/
│               ├── adapters/
│               ├── database/
│               ├── fragments/
│               ├── models/
│               ├── repository/
│               └── viewmodels/
```



6. Java Code (Activities) – Explanation

6.1 LoginActivity.java

Handles:

- User login
- Credential validation
- Navigation based on role

6.2 RegisterActivity.java

Users create accounts by providing:

- Username
- Password
- Role
- District

Stored in Room database.

6.3 MainActivity.java

Starting point of the application.

Displays:

- Event list
- Filters
- Navigation buttons

6.4 CreateEditEventActivity.java

For organizers to:

- Create new events
- Edit existing events

6.5 AdminPanelActivity.java

Admin can:

- View pending events
- Approve or reject them

6.6 EventListActivity.java

Displays all approved events using RecyclerView.

6.7 EventDetailActivity.java

Shows full information of an event.

7. XML Layouts – Explanation

activity_login.xml

- Username field
- Password field
- Login button

activity_main.xml

- Event list
- Navigation options based on role

activity_create_event.xml

Form for:

- Title
- Date
- Location
- District

activity_admin_panel.xml

List of pending events.

event_item.xml

Used inside RecyclerView to display:

- Title
 - District
 - Date
 - Favorite icon
-

8. Database / Storage

The system uses **Room Database**, with entities:

User Entity

- id
- username
- password
- role

Event Entity

- id
 - title
 - description
 - date
 - location
 - district
 - organizerId
 - approved
 - favorite
-

9. Application Flow

1. User opens the app
 2. Login/Registration
 3. Dashboard opens based on role
 4. Users view approved events
 5. Organizers create events
 6. Admin approves events
 7. Events visible to all users
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10. Implementation Steps

1. Create Room database and entities
 2. Implement MVVM (Model, ViewModel, Repository)
 3. Design UI screens
 4. Implement role-based authentication
 5. Implement CRUD for events
 6. Build admin approval module
 7. Add district filters and favorites
 8. Test all functionalities
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11. Installation & Running the App

- Open Android Studio
- Import the project
- Sync Gradle
- Run on emulator or real device

Minimum Requirements:

- Android 5.0+
 - 1 GB RAM
 - 50 MB free storage
-

12. Conclusion

The Local Event Hub application fulfills its objectives by providing a complete event management system with user, organizer, and admin roles. The application is simple, offline-ready, and scalable for future enhancements such as cloud storage, maps, or notifications.

Possible improvements:

- Firebase integration
 - Push notifications
 - Location-based event recommendations
-

13. Demo Credentials

Role	Username	Password
User	user1	pass123
Organizer	organizer1	pass123
Admin	admin	admin123
